The Efficient Frontier of Dollar-Cost Averaging

One-page summary — balancing expected wealth against terminal pain. Jaggernaut Wealth Services, June 2026.

The setup

A hypothetical investor earning at the 75th percentile of full-time income saves 20% of gross pay every month and dollar-cost-averages it into one portfolio over 1980–2026 (46 years, roughly age 21 to 67). 56 strategies — static allocations, age-based glides, and momentum/signal overlays — are each run in a fully taxable and a fully tax-free account, on the actual history plus 1,000 block-bootstrap simulated histories. Strategies are ranked on the **distribution** of outcomes, not the single realized path.

Three terms the whole analysis turns on

- **Median wealth** — rank the 1,000 final balances low to high and take the 501st (a single middle path, not an average): 500 histories ended poorer, 499 richer.
- **Terminal pain** — the 10th-percentile drawdown over the **final five years** before retirement, when the balance is largest and there is little time to recover. The report caps tolerable terminal pain at –30%, judging a deeper late-life loss likely unrecoverable.
- **Efficient** — a strategy is efficient if no other beats it on *both* median wealth and terminal pain; the rest are “dominated.” The efficient set within the –30% cap is the report’s shortlist.

Key findings

- Over 46 years the same contribution stream produced median final balances of about **$2.1M to $8.7M** depending on strategy and account.
- **More equity meant more wealth and deeper drawdowns together.** The highest-wealth mixes ($6–8M) also carried the deepest terminal drawdowns (–40% to –59%). No strategy had both the most wealth and the smallest drawdown.
- **Diversification cushioned the worst years.** In 2008, 2002 and 2022 a diversified stocks/gold/bonds mix fell far less than 60/40; 2022 (stocks and bonds down together) is the case gold helped most.
- **Taxes lowered wealth for every strategy** — the taxable account ended roughly 15–30% below the tax-free one. Hard rebalancing realizes taxable gains, so it ranked among the best in tax-free accounts and lower in taxable ones.
- **Only a handful were efficient** within the –30% cap: five in the taxable account, four in the tax-free; the rest were dominated.

The efficient set (not beaten on both wealth and terminal pain, and within the –30% cap)

Taxable account

Strategy	Median	Term. pain
TDF 90Stk/10Bnd→30Stk/70Bnd (hard)	$3.21M	–19%
All Weather, no rebal: 30Stk/40LongT/15Bnd/15Gld	$3.97M	–28%
All Weather, soft: 30Stk/40LongT/15Bnd/15Gld	$3.46M	–25%
Glide D, soft: 50Stk/25Gld/25Bnd→40Stk/30Gld/30Bnd→30Stk/30Gld/40Bnd	$3.57M	–27%
Permanent Portfolio, hard: 25Stk/25Gld/25LongT/25Cash	$2.10M	–18%

Tax-free account (Roth/401k)

Strategy	Median	Term. pain
TDF 90Stk/10Bnd→30Stk/70Bnd (hard)	$4.93M	–18%
60Stk/40Bnd (hard)	$5.48M	–28%
All Weather, no rebal: 30Stk/40LongT/15Bnd/15Gld	$5.07M	–27%
Permanent Portfolio, hard: 25Stk/25Gld/25LongT/25Cash	$3.00M	–16%

Three strategies appear in both sets: the 90Stk/10Bnd→30Stk/70Bnd target-date glide (hard), All Weather, and the Permanent Portfolio (hard). Both sets are unchanged under a second random seed.

What the report concludes

The best strategy by wealth per unit of terminal pain, in both accounts, is the hard-rebalanced 90Stk/10Bnd→30Stk/70Bnd glide — so “risky young, safe old” is supported. For that *high-equity* glide the hard version is specifically required: its soft variant never trims the early 90% equity and overshoots the cap (–33 to –34%). But hard rebalancing is not required to be efficient in general — the frontier within the cap also holds a lower-equity soft glide (Glide D) and All Weather run soft or with no rebalancing. The one big caveat: intermediate bonds — the dominant holding in the top glide — never had a severe drawdown in 1980–2026 (worst –19%), so the simulation cannot test the bond-heavy winner against a serious bond decline. An investor unwilling to hold 70% bonds near peak balance would instead take a diversified stocks/gold/bonds mix within the cap, splitting the defensive weight between gold and bonds rather than bonds alone.

Stk = US stocks · Bnd = intermediate bonds · LongT = long Treasuries · Gld = gold · Cash = short Treasuries. “Hard” sells to rebalance to target each year (realizing taxable gains); “soft” steers only new contributions and never sells; “no rebal” splits each contribution at target and lets winners run.

Source: The Efficient Frontier of Dollar Cost Averaging, Jaggernaut Wealth Services, LLC, June 2026. This is a backtest and thought experiment, not financial advice. Figures are nominal dollars from public data; the analysis relies on AI-written code and is spot-checked but not fully verified — reproduce any figure before relying on it.